CNN Project: YOLO / R-CNN Model Implementation
Last Date of Submission: 15 February 2026

Objective:
The objective of this project is to design, implement, fine-tune, and evaluate a Convolutional Neural Network (CNN)-based object detection model using YOLO or R-CNN.

Project Guidelines:
1. **Dataset Selection:**
Select a suitable object detection dataset from any of the following authentic repositories:
Primary Dataset Sources

- Kaggle Datasets: https://www.kaggle.com/datasets
- UCI Machine Learning Repository: https://archive.ics.uci.edu/
- Roboflow Universe : https://universe.roboflow.com/
- Google Dataset Search: https://datasetsearch.research.google.com/
- Open Images Dataset :https://storage.googleapis.com/openimages/web/index.html
- COCO Dataset : https://cocodataset.org/
- PASCAL VOC Dataset: http://host.robots.ox.ac.uk/pascal/VOC/
- ImageNet Dataset: https://www.image-net.org/

Dataset Requirements:

- The dataset must be suitable for object detection tasks.
- Annotation format conversion (YOLO / COCO / VOC format)

2. Model Selection:
Choose one of the following object detection models:

- YOLO (You Only Look Once)
- R-CNN (Region-Based Convolutional Neural Network)
  (Fast R-CNN / Faster R-CNN also acceptable)

4. **Results and Analysis:**
5. **Report Preparation** :
Submit an editable Microsoft Word file (.doc/.docx) including:

1. Dataset Description
2. Model Architecture
3. Workflow Diagram
4. Experimental Setup
5. Results (tables + graphs)
6. Analysis & Discussion

7. **Submission Instructions:**

- Upload the editable Word file to Google Classroom